

σ -ADDITIVE PROBABILITY ON ORTHOMODULAR LATTICES: A SURVEY AND AN OPEN PROBLEM

PATRICK S. MAHON

ABSTRACT. On a Boolean algebra, a finitely coherent probability extends to a countably additive measure by classical machinery (Carathéodory; Loomis–Sikorski; Stone). On a non-distributive orthomodular lattice (OML) this machinery fails. We survey the problem and isolate a precise open question, splitting it into two axes the Boolean case fuses: the *extension* axis (does a state extend to a charge on the dual?), a Horn–Tarski feasibility question settled negatively for $L(H)$; and the *descent* axis (does the charge concentrate on the physical points?). The combinatorial route fails — $\prod_n \text{MO}_2$ is concrete, σ -complete and non-Boolean yet *segregated* (non-distributivity confined to finite blocks) — because a faithful set representation forces the OML Boolean (Theorem 5.2). Reading “no hidden realisation” as not being a mixture of dispersion-free states makes a relational measure a *contextual* state (Lemma 5.3). The open question: *does a non-Boolean, σ -complete, concrete OML carry a σ -additive contextual state whose contextuality is σ -essential* — witnessed by no finite sub-OML? The finite case is inhabited (Wright’s pentagon), so only the σ -essential refinement remains, a compactness failure of CE type. Recent work on Dynkin-systems (Derr–Williamson [9]) settles it *negatively under a Polish/Borel-regularity hypothesis*, leaving the residue — whether a witness can be non-Polish-representable — a descriptive-set-theory question (Remark 5.1).

1. INTRODUCTION

All measurement is finite, yet the frameworks that organise empirical knowledge — probability theory, quantum mechanics — rest on infinite structures. The passage from finite observation to infinite structure is not automatic; it requires a commitment no finite body of evidence can compel. In probability theory the locus of that commitment is *countable additivity*:

$$A_1 \supseteq A_2 \supseteq \cdots, \quad \bigcap_n A_n = \emptyset \quad \implies \quad \mu(A_n) \rightarrow 0.$$

De Finetti [8] held that only finite additivity is empirically grounded; Kolmogorov [27] adopted σ -additivity as an axiom.

The same tension recurs once observations need not be jointly performable. An observation *context* — a set of measurements performable together — is Boolean; *pasting* contexts along their shared observations (the horizontal-sum / partial-Boolean-algebra construction [23]) returns:

$$\begin{aligned} \text{contexts mutually compatible} &\Rightarrow \text{Boolean algebra,} \\ \text{incompatible} &\Rightarrow \text{non-distributive OML.} \end{aligned}$$

The pasting of incompatible Boolean blocks is a non-distributive OML (Kalmbach [23]), and its failure to remain Boolean is exactly the contextuality obstruction of the partial-Boolean-algebra / sheaf account (Abramsky–Brandenburger [1]; Budroni–Morchio [2]). The OML is thus *forced* by the context structure, not postulated; for non-quantum examples the incompatibility must be operationally imposed, since classical observables on a common space jointly distribute (Kolmogorov) and stay Boolean. The closed subspaces of a Hilbert space H form the prototype $L(H)$, not the premise.

On a non-distributive OML, Carathéodory’s outer measure does not apply. For $L(H)$ it is rescued by Gleason’s theorem [16], with no general-OML analogue known. Non-distributivity *per se* does not

remove the two-valued homomorphisms the Boolean engine needs (MO_3 keeps them), but it removes the *guarantee*; for $L(H)$, $\dim \geq 3$, they fail outright (Kochen–Specker [26]). Whether enough survive is a question of *concreteness*, not non-distributivity: an OML is *concrete* (Definition 2.7) when it carries an order-determining set of two-valued states (Gudder [18]), and this is the qualifier on which the open problem turns.

The organising split. The question splits into two axes the Boolean case fuses: *extension* (does a state extend to a charge on $S_0(A)$? — settled, none does on $L(H)$) and *descent* (does the charge concentrate on the physical points? — open).

The relational standpoint. Following De Finetti [8] (probability without a presupposed sample space) and Birkhoff–von Neumann (propositions as a lattice, not subsets of a phase space), we start from the propositions and ask what their order alone determines — the measure-theoretic counterpart of pointless topology. The dual is then the McDonald–Bimbó space of *filters* (§2.3), and a positive answer would be a σ -additive probability theory on a non-distributive lattice, the non-Boolean analogue of localic measure theory (made precise in §5 as a *contextual* state; cf. [29]).

2. PRELIMINARIES

We collect the lattice-theoretic apparatus; all notions are standard.

2.1. Orthomodular lattices.

Definition 2.1 (Orthocomplemented lattice). A *bounded lattice* is a partially ordered set in which every pair a, b has a join $a \vee b$ and meet $a \wedge b$, with greatest element $\mathbf{1}$ and least element $\mathbf{0}$. An *orthocomplementation* is a map $a \mapsto a^\perp$ satisfying, for all a, b :

- (i) $a \wedge a^\perp = \mathbf{0}$ and $a \vee a^\perp = \mathbf{1}$ (complement),
- (ii) $a^{\perp\perp} = a$ (involution),
- (iii) $a \leq b \Rightarrow b^\perp \leq a^\perp$ (order-reversing).

Definition 2.2 (Incomparable; meet-zero; orthogonal). For elements a, b of an orthocomplemented lattice, three relations of increasing strength:

- *incomparable*: $a \not\leq b$ and $b \not\leq a$ — neither entails the other (an order condition);
- *meet-zero*: $a \wedge b = \mathbf{0}$ — no common refinement;
- *orthogonal*, written $a \perp b$: $a \leq b^\perp$ (equivalently $b \leq a^\perp$) — one entails the other’s negation, *decisively incompatible*.

Each implies the one above it ($a \perp b \Rightarrow a \wedge b = \mathbf{0} \Rightarrow$ incomparable, for $a, b \notin \{\mathbf{0}, \mathbf{1}\}$). For two *distinct atoms*, incomparable and meet-zero coincide — an atom has only $\mathbf{0}$ below it, so $a \wedge b \in \{\mathbf{0}, a\}$, and $a \wedge b = a$ would force $a \leq b$; orthogonality remains strictly stronger.

Remark 2.3 (Meet-zero versus orthogonal). The implication $a \perp b \Rightarrow a \wedge b = \mathbf{0}$ reverses in every Boolean algebra, but fails once the lattice is non-distributive: there meet-zero is strictly weaker than orthogonality. The distinctions developed below all turn on this separation.

Definition 2.4 (Orthomodular lattice). An orthocomplemented lattice A is *orthomodular* if

$$a \leq b \implies b = a \vee (a^\perp \wedge b). \quad (1)$$

A *Boolean algebra* is the special case where $\wedge$ distributes over $\vee$; every Boolean algebra is orthomodular, not conversely. The motivating non-Boolean example is $L(H)$, the closed subspaces of a Hilbert space with $a^\perp$ the orthogonal complement.

Definition 2.5 (Completeness notions). Two countable-completeness conditions on an OML A are distinguished throughout, the second strictly weaker. Here $\bigvee_n a_n$ denotes the *join* (least upper bound) of $\{a_n\}$ in the order of A , unique when it exists; “ $\exists \bigvee_n a_n \in A$ ” asserts that it does.

- A is σ -complete if

$$\forall \{a_n\}_{n \in \mathbb{N}} \subseteq A : \quad \exists \bigvee_n a_n \in A$$

(equivalently every countable family has a meet, by $\bigwedge_n a_n = (\bigvee_n a_n^\perp)^\perp$);

- A is σ -orthocomplete if

$$\forall \{a_n\}_{n \in \mathbb{N}} \subseteq A \ (a_i \perp a_j \text{ for } i \neq j) : \quad \exists \bigvee_n a_n \in A.$$

The orthogonal families are a subclass of all countable families, so σ -completeness quantifies over a strictly larger domain and is the stronger hypothesis: σ -completeness $\Rightarrow$ σ -orthocompleteness. Only the weaker condition is needed below. For a state s , σ -additivity is the requirement that

$$s\left(\bigvee_n a_n\right) = \sum_n s(a_n)$$

for every countable pairwise-orthogonal family $\{a_n\}$. Its left side invokes $\bigvee_n a_n$ *only* for orthogonal families — so σ -orthocompleteness is exactly the hypothesis that makes it well-posed. Full σ -completeness is the stronger condition manipulated by the Loomis–Sikorski representation (§2.4).

Example 2.6 (MO_n , and the gap made concrete). MO_n is the lattice with $\mathbf{0}$, $\mathbf{1}$, and n complementary pairs of incomparable atoms, all sharing only $\mathbf{0}$ and $\mathbf{1}$. The geometric model is n lines through the origin: each line a is orthogonal to its perpendicular $a^\perp$, while two atoms from distinct pairs are non-perpendicular lines. It is orthomodular and non-distributive ($n \geq 2$): two atoms drawn from *distinct* complementary pairs have

$$a \wedge b = \mathbf{0} \quad (\text{meet only at the origin}) \quad \text{yet} \quad a \not\perp b,$$

so meet-zero $\neq$ orthogonal already here. MO_3 is the smallest non-distributive OML.

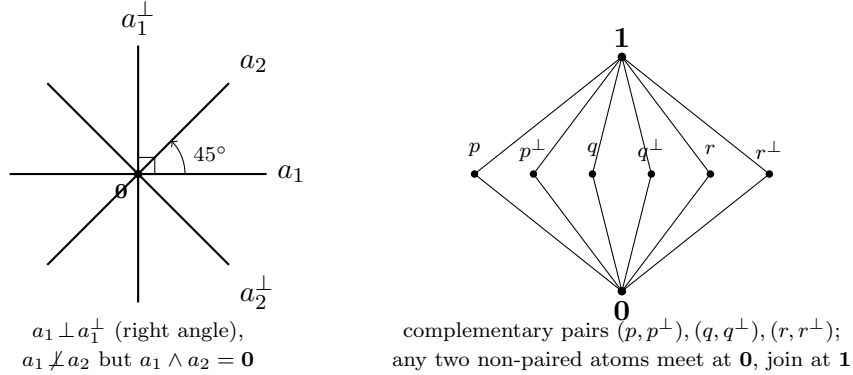


FIGURE 1. MO_n illustrated. *Left:* the geometric model for MO_2 — four atoms as lines through the origin in $\mathbb{R}^2$. The pair $a_1, a_1^\perp$ meets at a right angle (orthogonal), whereas a_1, a_2 meet at 45° : their lattice meet is still $\mathbf{0}$ (only the origin in common), yet they are *not* orthogonal — the meet-zero $\neq$ orthogonal gap. *Right:* the Hasse diagram of MO_3 , the smallest non-distributive OML: three complementary pairs of incomparable atoms between $\mathbf{0}$ and $\mathbf{1}$.

Definition 2.7 (OMP; order-determining states; concrete logic). An *orthomodular poset* (OMP) weakens Definition 2.4: joins $a \vee b$ are required only for *orthogonal* pairs. A set S of states on A is *order-determining* if it separates the order, i.e.

$$(s(a) \leq s(b) \text{ for all } s \in S) \implies a \leq b.$$

An OMP is a *concrete logic* (equivalently *set-representable*) if it embeds into $(\mathcal{P}(X), \subseteq, X \setminus (-))$ for some set X , preserving existing orthogonal joins — i.e. its propositions are modelled as actual *sets*

of *outcomes*. Concreteness is equivalent to admitting an order-determining set of *two-valued* states (Gudder [18]), and it is the property that separates the open frontier from the settled cases (§5).

Remark 2.8 (Concreteness does not close the gap). A set-representable OMP needs only closure under complement and *disjoint* union [4]; intersection-closure would force a field of sets, hence Boolean. The lattice meet $a \wedge b$ is the greatest L -member inside $A \cap B$, so

$$\underbrace{A \cap B = \emptyset}_{\text{orthogonal}} \implies \underbrace{a \wedge b = \mathbf{0}}_{\text{meet-zero}}, \quad \text{but not conversely.}$$

Concreteness does not force the converse: MO_3 is itself concrete (Definition 2.7; Gudder [18]), so the paradigmatic meet-zero $\neq$ orthogonal example is already a concrete logic. What closes the gap is a richness/atomicity condition, not concreteness:

$$a \wedge b = \mathbf{0} \implies a \perp b,$$

equivalently: every nonempty $A \cap B$ contains a nonzero L -member — the abstract form being Tkadlec’s *Boolean orthoposet* condition [39] (which does *not* mean Boolean *algebra*).

2.2. The state/meet-additive/charge ladder.

Definition 2.9 (State; meet-additive state; charge-extendible). Let A be an OML. A *state* is a map $s : A \rightarrow [0, 1]$ with $s(\mathbf{1}) = 1$ that is additive on *orthogonal* pairs: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$; that is, a probability assignment whose additivity is guaranteed only on decisively incompatible propositions. Three strengthenings, differing in which pairs additivity is demanded on and strictly increasing in general (Example 2.10 on MO_3):

- (1) *State*: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$.
- (2) *Meet-additive*: $s(a \vee b) = s(a) + s(b)$ whenever $a \wedge b = \mathbf{0}$ (binary, strictly stronger — it reaches the gap pairs of Definition 2.2). We avoid the word “valuation” here: in lattice theory it denotes the modular function $v(a) + v(b) = v(a \wedge b) + v(a \vee b)$, a different condition.
- (3) *Charge-extendible*: $\sum_i s(a_i) \leq 1$ for every pairwise-meet-zero family $\{a_i\}$ ($a_i \wedge a_j = \mathbf{0}$, $i \neq j$; strictly stronger again). This is the condition necessary for extension to a charge on the dual space (Definition 2.13) and the *extension* condition of §3.

Example 2.10 (The ladder is strict). On MO_3 with complementary pairs $(p, p^\perp), (q, q^\perp), (r, r^\perp)$, the maximally mixed $s \equiv \frac{1}{2}$ is meet-additive (clears (2)) yet fails (3) at $\{p, q, r\}$: $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} > 1$. So no state on MO_3 is charge-extendible.

Definition 2.11 (Normal and singular states on $L(H)$). On $A = L(H)$ a state s is

- *normal* if it is completely additive on every orthogonal family of projections,

$$s\left(\bigvee_{i \in I} p_i\right) = \sum_{i \in I} s(p_i) \quad \text{for all (not only countable) orthogonal } \{p_i\},$$

equivalently $s(\cdot) = \text{tr}(\rho \cdot)$ for a density operator ρ (Gleason [16], $\dim \geq 3$);

- *singular* (purely) if

$$s(e) = 0 \quad \text{for every finite-rank projection } e.$$

Every state decomposes into a normal and a singular part (Takesaki); the two classes are exactly those distinguished by the arguments of §3.

2.3. The McDonald–Bimbó dual space.

Definition 2.12 (Dual space; representation map). The McDonald–Bimbó duality [30] assigns to an OML A a compact space

$$S_0(A) = (F(A), \subseteq, \perp_A, P(A), T),$$

where $F(A)$ is the set of *filters* of A , ordered by $\subseteq$ and carrying an orthogonality relation $\perp_A$ and a Stone topology T , and $P(A) = \{\uparrow p : p \in A\} \subseteq F(A)$, with $\uparrow p = \{a \in A : a \geq p\}$, is the set of *principal* filters — the “physical points,” the realisation datum (canonical, unlike the non-canonical pure points of the Boolean case). For $a \in A$ set $h(a) = \{x \in F(A) : a \in x\}$; then h is an OML isomorphism onto the $\perp$ -stable clopens $\text{CO}(S_0(A))^\dagger$. Define $\mu(h(a)) = s(a)$. Here $h(a)$ is the set of points at which a holds and μ is the state transported onto those sets; the open problem is whether μ is preserved under countable operations on them.

Definition 2.13 (Charge). A *charge* on $S_0(A)$ is a map $\nu : \text{CO}(S_0(A)) \rightarrow [0, 1]$ on the *full* Boolean algebra of clopen subsets (not merely the $\perp$ -stable ones in the image of h), with

$$\nu(S_0(A)) = 1, \quad \nu(U \cup V) = \nu(U) + \nu(V) \text{ whenever } U \cap V = \emptyset.$$

A state s *extends to a charge* if some such ν satisfies $\nu(h(a)) = s(a)$ for all $a \in A$. Because h preserves meets, a pairwise-meet-zero family in A maps to disjoint clopens, so a charge necessarily satisfies $\sum_i s(a_i) \leq 1$ over such families — the charge-extendibility condition of Definition 2.9(3).

Remark 2.14 (The duality is finitary). The representation map h satisfies

$$\begin{aligned} h\left(\bigvee_{i \leq n} a_i\right) &= \left(\bigcup_{i \leq n} h(a_i)\right)^{\perp\perp} \quad (\text{finite}), \\ h\left(\bigvee_n a_n\right) &\neq \left(\bigcup_n h(a_n)\right)^{\perp\perp} \quad (\text{countable}) : \end{aligned}$$

finitary OML homomorphisms need not preserve infinite suprema. This is the central obstruction the open problem of §5 confronts, and it is self-dual: since $a \mapsto a^\perp$ is an order anti-isomorphism, $\bigwedge_n a_n = (\bigvee_n a_n^\perp)^\perp$, so the failure for countable meets is the De Morgan dual of the failure for countable joins. The rival duality of Cannon–Döring [5] is also finitary and carries no measure; we use McDonald–Bimbó because it carries $P(A)$ as explicit structure.

2.4. The Boolean engine: Stone duality. In the Boolean case the descent equivalence

$$\sigma\text{-additive} \iff \text{the measure concentrates on the physical points}$$

is proved through *Stone duality*, in four steps [29]:

- (S1) a Boolean event algebra B embeds in the clopen algebra $\text{CO}(\text{St } B)$ of its Stone space;
- (S2) a charge on B extends to a Baire measure on $\text{St } B$ (Carathéodory);
- (S3) σ -additivity $\iff$ the measure vanishes on meagre Baire sets (Rao–Rao [36]);
- (S4) meagre-null $\iff$ the measure lives on the ultrafilters closed under countable intersection = the realised points.

Steps (S3)–(S4) are the measure-theoretic face of the *Loomis–Sikorski* representation,

$$\sigma\text{-complete Boolean algebra} \cong (\sigma\text{-tribe of sets}) / (\sigma\text{-ideal}),$$

the countable-join bookkeeping that drives the equivalence. The contrast with the OML case is exact: the Boolean engine (Stone duality together with the Loomis–Sikorski representation) has *no* known OML analogue (§4.3). Supplying such an engine, or proving none exists, is the open problem of §5.

Remark 2.15 (Why the bridge fails: meet-closed but not join-prime). A “realised” point x carries two conditions, which Rao–Rao’s “vanishes on meagre sets” makes coincide in the Boolean case:

- (M) closed under countable meets;
- (J) $x \ni \bigvee_n a_n \Rightarrow x \ni a_n$ for some n (off the join-defect).

Non-distributivity separates them. For a principal filter $x = \uparrow p$, condition (M) always holds, whereas (J) can fail: $p \leq \bigvee_n a_n$ does not force $p \leq a_n$ for some n . In $L(H)$ this is witnessed by $a_n = \text{span}(e_n)$ and $p = \text{span}(e_1 + e_2)$:

$$p \leq \bigvee_n a_n = H, \quad \text{yet} \quad p \not\leq a_n \text{ for every } n.$$

So $\uparrow p$ is a realised point lying inside the join-defect $D = h(\bigvee_n a_n) \setminus \bigcup_n h(a_n)$. Topologically $\uparrow p$ is *isolated* in $S_0(L(H))$, hence:

$$D \text{ non-meagre (OML)} \quad \text{vs.} \quad D \text{ nowhere dense (Boolean).}$$

This is the finitary obstruction of Remark 2.14 on the dual: the category route (clopens modulo the meagre ideal) fails on the same D as the measure route, and reduces to the MacNeille completion of §4.3(ii).

3. EXTENSION AND DESCENT

The central question — *when does a state s on an OML A extend to a σ -additive measure on $S_0(A)$?* — splits into two *independent* axes that the Boolean case fuses.

Extension.:

Does s extend to a finitely additive charge on the *full* Boolean clopen algebra of $S_0(A)$? This is governed by the meet-zero/orthogonal gap (§2.1) and is *settled*: it is a Horn–Tarski/Pitowsky feasibility question, a decidable linear program in the finite case, and can fail for every state.

Descent.:

Once extended, does the charge concentrate on the physical points $P(A)$? This is governed by σ -additivity — it needs the missing engine (§2.4) — and is *open*.

The extension axis is the abstract form of a familiar question: extending a finitely-additive assignment on incompatible propositions to a single joint distribution is exactly the feasibility problem behind the Bell inequalities (Fine [13]: a Bell model exists iff the correlations admit a joint distribution; Pitowsky [33]: the Bell inequalities are the facets of a correlation polytope), and Bell and Kochen–Specker non-classicality are jointly an extension problem for partial Boolean algebras (Budroni–Morchio [2], with the necessary-and-sufficient conditions of Horn–Tarski [22]). It is also the marginal problem of probability theory — extending consistent marginals to a joint law (Kellerer [24]) — and the coherence/extendability question of imprecise probability (Walley [42]); the equivalence of these formulations to the quantum-logical one is made precise by Derr–Williamson [9]. What is distinctive here is only that the extension is sought against the dual $S_0(A)$ rather than within a fixed scenario; the obstruction is the same finitary one, and it is settled. Two observations pin it down.

Proposition 3.1 (Finite case). *For finite A the extension condition is a decidable linear program (Horn–Tarski [22] / Pitowsky correlation-polytope feasibility [33]; Budroni–Morchio [2] for the partial-Boolean-algebra formulation); since h preserves meets, meet-zero elements map to disjoint clopens and a charge must satisfy $\sum_i s(a_i) \leq 1$ over pairwise-meet-zero families. It can fail for every state (MO₃, Example 2.10).*

Proposition 3.2 (Infinite $L(H)$, all states). *Let $\dim H = \infty$. No state on $L(H)$ — normal or singular — extends to a charge on $S_0(L(H))$.*

Proof sketch. Embed MO₂ in $L(H)$ with *infinite-dimensional* atoms: writing $H = H_0 \otimes \mathbb{C}^2$ ($\dim H_0 = \infty$), the four subspaces $H_0 \otimes \mathbb{C}e$, $H_0 \otimes \mathbb{C}f$, $H_0 \otimes \mathbb{C}(e+f)$, $H_0 \otimes \mathbb{C}(e-f)$ are pairwise meet-zero with $a_i \vee a_i^\perp = H$. Orthoadditivity forces $\sum_i s(a_i) = 2$ for every state, while disjointness of the images under the meet-preserving h forces $\sum_i s(a_i) \leq 1$ for any charge — a state-independent collision. The construction is finite ($n = 4$), so σ -completeness plays no role; the infinite-dimensional atoms are what carry the obstruction past the finite-rank states to the singular ones (where every finite-rank projection has measure zero), closing the axis completely. $\square$

4. WHAT IS KNOWN

The extension axis is settled — on $L(H)$ no state extends (Proposition 3.2); the descent axis turns on whether the Boolean engine of §2.4 has an OML analogue. Three things bear on that:

1. the Boolean benchmark such an analogue must match (§4.1);
2. the partial OML results, positive and obstructive (§4.2);
3. the routes already known to be blocked (§4.3).

4.1. The Boolean benchmark. The conditions an OML engine would have to reproduce. The Kelley–Vladimirov–Pták characterisation (Fremlin [14], Thm 391D) is complete: a Boolean algebra carries a strictly positive σ -additive measure iff it satisfies all three conditions below — each of which loses its meaning off the distributive world.

Boolean (KVP) condition	OML analogue
Dedekind σ -completeness	exists (σ -orthocompleteness)
weak (σ, ∞) -distributivity	none — relies on $\wedge$ distributing over $\vee$
chargeability	none — no structural characterisation on OMLs

4.2. Known results for general OMLs. Positive (require structure rich enough to approximate Hilbert-space geometry):

- Gleason [16] — on $L(H)$ for a *separable* Hilbert space of $\dim \geq 3$, every countably additive measure on the projection lattice has the trace form $s(\cdot) = \text{tr}(\rho \cdot)$: σ -additivity *on the lattice*.
- Bunce–Wright [3] — the projection-lattice (JBW) form; Chetcuti–Dvurečenskij [6].

Obstructive (Boolean machinery cannot be transplanted):

- Pták–Pulmannová [35] — a *finitary* condition (*subadditive*-unitality $s(a \vee b) \leq s(a) + s(b)$, no countable joins) already collapses the *lattice* to Boolean (their Thm 1); the *OMposet* version fails (Müller’s set-representable counterexample [31]).
- Navara [32] — regularity does not force σ -additivity; with Rüttimann’s decomposition of states into completely additive and purely finitely additive parts [37], this shows the finitely-additive/ σ -additive gap is structurally serious on quantum logics, not a routine strengthening — part of why the open problem is hard. Pták [34] — exotic state spaces via Greechie pasting.

4.3. Three routes to a σ -OML engine. The descent argument needs an OML analogue of the Boolean engine — a σ -Stone duality, equivalently a Loomis–Sikorski representation. Routes (i),(ii) are closed by *theorem*; route (iii) is open in the strong sense (no construction known, none ruled out), and is the Open Problem of §5.

- (i) **RDP / effect-algebra.** Loomis–Sikorski holds for σ -MV algebras [11] and for monotone σ -complete effect algebras *with* the Riesz Decomposition Property [12]; but a lattice effect algebra has RDP iff it is MV, and $\text{OML} \cap \text{MV} = \text{Boolean}$ (Kalmbach [23]): $\text{OML} + \text{RDP} \iff \text{Boolean}$. The *sharp-skeleton* evasion (represent the OML as the sharp elements of an ambient RDP effect algebra) also closes: in an RDP effect algebra the sharp elements coincide with the centre, hence are Boolean (Jenča 2001, Cor. 4.3, via Greechie–Foulis–Pulmannová [17]).
- (ii) **MacNeille completion.** The MacNeille completion of an OML need not be orthomodular (Harding [19]), so one cannot complete to absorb countable joins; the topological version (clopens of $S_0(A)$ modulo the meagre ideal) is the same completion and collapses non-distributivity (Remark 2.15).
- (iii) **σ -Stone duality.** None is known, none ruled out: the existing dualities are finitary (McDonald–Bimbó, Remark 2.14) or equational with no set/tribe representation (Freytes [15]), but no theorem forbids a countable-join-preserving one. This is route (c) of the faithfulness

trichotomy (§5.1): a representation faithful on countable *orthogonal* joins without forcing distributivity.

4.4. Prior art on point-free quantum probability. Every existing point-free or directed-context construction fails at least one of the two requirements — σ -additivity and a natively non-distributive structure — as tabulated below.

Gleason tradition:

Varadarajan [40], Kalmbach [23], Pták–Pulmannová [35]: σ -additive on a non-distributive OML, but real-valued on a fixed lattice — not point-free.

Döring 2009:

[10] point-free on the spectral presheaf over a directed poset — but provably only *finitely additive*, on a distributive Heyting algebra.

Bohrification (HLS):

[20, 21] point-free and the internal valuation *is* σ -additive (Scott-continuous) — but factors through an internally *distributive* locale, σ -additive only per commutative context. Non-distributivity is tamed, not carried natively.

Localic valuations:

Vickers [41], Simpson [38], Coquand–Spitters [7]: point-free σ -additive, but on *frames* — distributive by definition.

OML Stone dualities:

McDonald–Bimbó [30] and Cannon–Döring [5]: purely structural, no measure, no σ -version.

The open case is their conjunction, met by no existing construction:

point-free $\times$ σ -additive $\times$ natively non-distributive $\times$ directed-system-built.

Remark 4.1 (The boundary of the negative results). Two qualifications delimit what is ruled out.

- (i): **No impossibility theorem covers the concrete class.** The known no-go results are finite or two-valued, and the one positive occupant is non-concrete:

$$L(H) \text{ (dim} \geq 3) : \quad \text{no two-valued states at all (Kochen–Specker [26]),}$$

a fortiori none order-determining. Concreteness is the qualifier on which the open frontier turns.

- (ii): **No σ -Loomis–Sikorski exists** (route (c), §5.1): the dualities take infinite joins as a closure, not a union (Remark 2.14). Not to be overstated, though: Gudder’s concrete logics [18] *are* set-representable non-Boolean orthoposets — but as point-ful posets, not via a σ -Loomis–Sikorski theorem.

4.5. The topos approach. The topos / Bohrification programme is the one machinery supplying a point-free *and* σ -additive valuation on a quantum logic, so it is the natural candidate — but it takes route (b) of the trichotomy (§5.1): in both principal constructions the valuation lives on a *distributive* object, its countable additivity routed through per-context Boolean blocks, hence sidesteps rather than settles route (c).

	Bohrification (HLS)	Döring–Isham
measure on	internal Gelfand spectrum $\Sigma(A)$, a compact regular frame	clopen subobjects $\text{Sub}_{cl}(\Sigma)$, a Heyting algebra
why distributive	a frame: $\bigvee$ distributes [21, Def. 6.11]	a locale, not a σ -algebra [10]
how Proj enters	Thm 6.19 bijection; the measure restricts to a σ -Boolean morphism on each Boolean sublattice (Def. 6.17(a))	daseinisation, which “cannot preserve both [joins and meets]” [43]
additivity left	block-by-block, glued by naturality; never crosses a non-commuting join	only <i>finitely</i> additive; σ per-block, normal states only

Two route-(b) details specific to the topos case: HLS confines all additivity inside Boolean contexts (the antithesis of σ -essential contextuality), and its non-distributive object $\text{Proj}(A)$ is the non-concrete Gleason / $L(H)$ lattice (Rmk 4.1), not the concrete class in question — so of the four requirements (point-free, σ -additive, natively non-distributive, directed-system-built) it misses exactly native non-distributivity.

5. THE OPEN PROBLEM

Open Problem. Does there exist a concrete, σ -complete, non-Boolean OML L and a σ -additive state w on L such that

$$\begin{aligned} (\text{contextual}) \quad & w \notin \overline{\text{conv}}(S_{\text{df}}^\sigma); \\ (\sigma\text{-essential}) \quad & w|_F \in \overline{\text{conv}}(S_{\text{df}}(F)) \text{ for every finite sub-OML } F \subseteq L \end{aligned}$$

That is, w has no global hidden joint over the dispersion-free states, yet its restriction to every finite sub-OML does. Equivalently: can one prove no such (L, w) exists?

Remark 5.1 (Relation to Derr–Williamson 2023). The problem above is not plainly open: it must be read against Derr–Williamson [9], who study coherent probabilities on *pre-Dynkin-systems* (concrete orthomodular posets) and *Dynkin-systems* (concrete σ -complete OMLs) from the imprecise-probability side. Three distinct levels of statement are involved, and should not be conflated:

- (a) a finite-additive extendability criterion (their Theorem 4.5, a Horn–Tarski feasibility condition) matching “non-contextual on every finite sub-OML”;
- (b) a *countably additive* extension to the ambient Borel σ -algebra (their Theorem D.6), valid *under the hypotheses* that the OML is realised as a Dynkin-system D_σ of Borel sets of a Polish space Ω with $\sigma(D_\sigma) = \text{Borel}(\Omega)$ and inner-regular σ -blocks — a result they obtain by specialising Maharam’s extension theorem [28, Thm 8.1] (Hausdorff base, inner-regular measures); and
- (c) the inference from (b) to non-contextuality of w , below, which is the survey’s own step.

The link to the present formulation is one-directional but in the direction that matters. A point $\omega \in \Omega$ gives the evaluation δ_ω , which is a two-valued and (since $A_n \downarrow \emptyset \Rightarrow \omega \notin A_n$ eventually) σ -additive dispersion-free state; a countably additive ν on $\text{Borel}(\Omega)$ with $\nu|_{D_\sigma} = w$ exhibits w as a barycentre,

$$w(A) = \int_{\Omega} \delta_\omega(A) d\nu(\omega) \implies w \in \overline{\text{conv}}(S_{\text{df}}^\sigma).$$

Thus

$$\sigma\text{-extendable} \implies \text{non-contextual},$$

and through Theorem D.6 the implication “finitely non-contextual everywhere $\Rightarrow$ globally non-contextual” holds *whenever the OML admits such a Polish Borel realisation*: under that hypothesis the σ -essential cell is empty, and no witness exists. (The converse, non-contextual $\Rightarrow \sigma$ -extendable, does not hold — the dispersion-free states properly contain the point evaluations δ_ω — but it is not needed for the negative conclusion.)

The residue is then a single condition, the representability the whole problem turns on: can a σ -essential witness exist that admits *no* Polish Borel realisation (S_{df}^σ non-Polish / analytic-but-not-Borel)? This is a descriptive-set-theory question, equivalent to the absence of a σ -Loomis–Sikorski representation (§4.3); the off-centre/non-band constraints of §6 are its lattice-side counterpart. One caution concerns the σ -version of the (A)/(B) point-space equivocation (Remark 5.6): Derr–Williamson’s Polish hypothesis is on the underlying *point set* Ω , and obtaining such an Ω for an abstract concrete σ -complete OML is itself the missing engine, not a free input.

Here “relational” means *no hidden realisation*: the measure is not recoverable from an auxiliary space of definite states behind the propositions. Three point-spaces must be kept distinct: the *canonical* dual $P(A) = \{\uparrow p\}$ (one principal filter per element, §2.3); the *dispersion-free states* (the hidden-variable points); and the *external (A)-points* (Gleason’s Hilbert rays, Remark 5.6). $P(A)$ is *permitted*; the objection is only to a measure built on an illicitly introduced space — the (A)-points, equivalently a global hidden-variable joint over the dispersion-free states. By Lemma 5.3 a state has a hidden realisation iff it is a mixture of dispersion-free states, so a relational measure is precisely a *contextual* state. This is a condition on the dispersion-free states, not on $P(A)$: a witness is point-rich on $P(A)$ yet has no global dispersion-free joint.

5.1. Reduction to contextuality.

The universal floor. The naive route — realise the lattice as honest sets so that a measure becomes a measure on a space — is closed for every non-Boolean OML.

Theorem 5.2 (Universal floor). *A faithful set representation of an OML L — an injective $h : L \rightarrow \mathcal{P}(\Omega)$ with $h(a \wedge b) = h(a) \cap h(b)$, $h(a \vee b) = h(a) \cup h(b)$, $h(a^\perp) = \Omega \setminus h(a)$ (a σ -tribe / Loomis–Sikorski representation when σ -joins are preserved) — exists only if L is Boolean.*

Proof sketch. Set operations $\cap, \cup, ^c$ on $\mathcal{P}(\Omega)$ distribute, so the image $h(L)$ is a distributive sublattice; h injective and lattice-preserving makes L isomorphic to a distributive ortholattice, i.e. Boolean. $\square$

The universal floor (Theorem 5.2) fixes *where* a σ -engine may relinquish faithfulness, a trichotomy organising §§4.3–4.4: **(a)** all $\wedge, \vee$ as $\cap, \cup$ forces Boolean (Theorem 5.2), and is *excluded*; **(b)** all joins but in a *distributive* ambient object (frame/topos) is σ -additive only per Boolean context (§4.5), and *sidesteps* rather than settles the question; **(c)** only countable *orthogonal* joins, faithful to order and $\perp$ but not to the distributivity-forcing meets — a concrete logic, exactly what σ -additivity invokes — is *the remaining open route* (§4.3(iii)), elaborated next. Route (c) is the candidate for the missing σ -duality — the contravariant equivalence one would want, between concrete σ -complete OMLs and a category of σ -spaces whose points are the dispersion-free states. The universal floor fixes what is *forced* on the right-side object: non-distributivity cannot live in the sets (faithful $\wedge, \vee \mapsto \cap, \cup$ forces Boolean, Theorem 5.2), so it sits in the meet being a sub-intersection under the Gudder/concrete-logic functor (orthogonal joins as disjoint unions, set-complement); the descent space is S_{df}^σ , *not* the principal points $P(A)$ (permitted; Remark 5.6); the field of sets is σ -generated, not Stone clopen. Under that functor the *representation* is settled — a concrete σ -Dynkin-system does not collapse to Boolean (Remark 2.8; $\prod_n \text{MO}_2$ inhabits it) — but this is a *definitional pin*, not a theorem: fixing the target to σ -Dynkin-systems makes it hold by the chosen functor. The entire weight thus rests on the remaining clause, the *measure descent* — whether a σ -additive state is a mixture of the S_{df}^σ , off the regular-realisable case (Remark 5.1). So a concrete OML has an order-determining family of dispersion-free states; the one open question is whether every state is a mixture.

The reduction. The following equivalence reduces “no hidden realisation” to a spanning condition on the dispersion-free states. Its content is not new — for a Bell scenario it is Fine’s theorem [13] (a joint distribution exists iff a deterministic hidden-variable model does), and the convex-geometric form is Pitowsky’s correlation-polytope picture [33]; the formulation for a general (concrete) OML, with dispersion-free states as the deterministic vertices, is the natural lift, in the lineage of the sheaf-theoretic account of contextuality [1].

Lemma 5.3 (Relational = contextual). *Let L be a concrete OML with dispersion-free state space S_{df} , and w a state on L . Then w admits a hidden realisation — a probability measure on a space of global two-valued valuations of L inducing w — if and only if $w \in \overline{\text{conv}}(S_{\text{df}})$. We call w contextual when it is not.*

Proof sketch. A global two-valued valuation of L is exactly a dispersion-free state, so a hidden realisation is a probability measure μ on S_{df} with $w(a) = \int \omega(a) d\mu(\omega)$ for all a — i.e. w is the barycentre of μ . The barycentres of probability measures on the (weak-* compact) set S_{df} are exactly $\overline{\text{conv}}(S_{\text{df}})$. The handle is *spanning*, not simplicity: non-uniqueness of μ is irrelevant; only its *existence* matters. $\square$

The object sought is therefore a concrete σ -complete OML with a σ -additive state outside the dispersion-free hull (Figures 2 and 3).

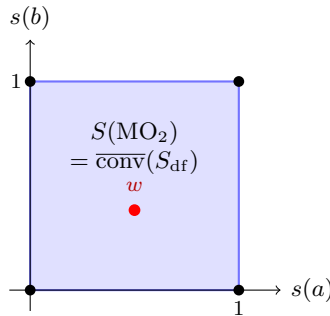


FIGURE 2. MO_2 : the state space is the square $[0, 1]^2$ (a state is fixed by $(s(a), s(b))$), and the four dispersion-free states are its corners, so $\overline{\text{conv}}(S_{\text{df}})$ is the whole square. Every state w is a mixture of corners — none is contextual. Thus non-distributivity alone does not produce contextuality (contrast Figure 3): it requires a loop, not merely incompatible elements.

Why σ -essential. The finite version of this is already inhabited. Wright’s pentagon [44] is a finite, concrete OML — a Greechie loop of five three-atom blocks, hence a lattice — with an order-determining family of dispersion-free states, yet it carries a non-classical $\{0, \frac{1}{2}\}$ -valued state lying outside their closed convex hull. (The five intertwining atoms form a 5-cycle in which consecutive atoms share a block, so any dispersion-free state, taking at most one of each adjacent pair to 1, sets at most 2 of the 5 to 1; their hull has vertex-sum ≤ 2 , whereas the $\{0, \frac{1}{2}\}$ state — a legitimate state, each block summing to 1 — has vertex-sum $\frac{5}{2}$. This is the KCBS inequality [25].) So “concrete OML with a contextual state” is a 1978 fact, and the general spanning statement is false. What Wright does *not* supply is contextuality that requires the countable structure: the pentagon’s is witnessed by a finite sub-OML (itself). The such a state must therefore be σ -essential — contextual in the σ -complete whole but in no finite sub-OML. This is not a device to evade Wright: “every finite piece has a global section, the countable whole has none” is a *compactness failure* of exactly the type by which countable additivity is not first-order axiomatizable over finitely additive coherence (see Remark 5.4).

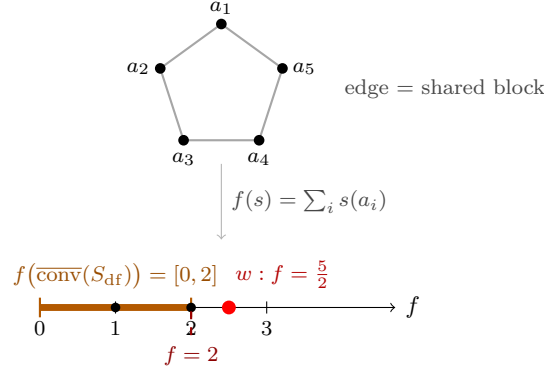


FIGURE 3. Wright's pentagon (a Greechie 5-loop; left), and the same hull-membership question read off a separating functional (right). The state space is 5-dimensional and undrawable, but $f(s) = \sum_i s(a_i)$ projects it faithfully to $\mathbb{R}$: every dispersion-free state has $f \leq 2$ (the odd loop admits no two adjacent atoms valued 1), so $f(\overline{\text{conv}}(S_{\text{df}})) = [0, 2]$. The non-classical state $w \equiv \frac{1}{2}$ has $f = \frac{5}{2}$, past the face $f = 2$, hence outside the hull — contextual. (Here f is a *chosen* separating functional for this w ; by the separating-hyperplane theorem a state is contextual iff *some* linear functional sends it past the hull's image.)

Status. No witness has been constructed: the product $\prod_n \text{MO}_2$ fails (its contextual states are only finitely additive — diffuse states on its central $P(\mathbb{N})$, which σ -additivity forces to concentrate on points, killing the contextuality, §5.2); Wright is finite; no other construction is known. On the negative side, the cell is now known to be *empty for every OML admitting a Polish Borel point-realisation* (Derr–Williamson, Remark 5.1). What remains genuinely open is therefore the *non-representable* case: a witness, if one exists, must admit no such realisation — which is precisely the absence of a σ -Loomis–Sikorski representation (§4.3), a descriptive-set-theory question.

Remark 5.4 (The compactness-failure analogy). The σ -essential requirement places the open problem alongside a known Boolean phenomenon. Over a Boolean algebra, countable additivity of a finitely additive charge is not first-order axiomatizable: an ultraproduct of σ -additive charges can be finitely additive but not σ -additive, so every finite test is passed while the global property fails [29]. The σ -essential contextual state is the non-distributive analogue, but with one sharpening the Boolean case obscures: the obstruction is *finitely detected but globally caused*. Each finite sub-OML is classically interpretable — its restriction lies in the full local hull $\overline{\text{conv}}(S_{\text{df}}(F))$ — yet contextuality is detected already at a single finite F , against the strictly smaller hull $\overline{\text{conv}}(\{s|_F : s \in S_{\text{df}}^\sigma\})$ of restrictions of the *global* σ -states. The gap between the two is what the countable whole supplies; it is not a property that emerges only in the limit (a vanishing-in-the-limit margin would be failure of σ -additivity itself, the segregated $\prod_n \text{MO}_2$ case, not a witness).

5.2. The combinatorial route. One might seek the object among concrete σ -orthocomplete OMLs, on the grounds that a state's σ -additivity invokes $\bigvee_n a_n$ only for orthogonal families (Definition 2.5). This route is settled, and does not produce the object. The property that would make σ -additivity genuinely exceed finite additivity on a concrete non-Boolean lattice is the interleaving condition

$$(\star) \quad \exists p, \exists \{a_n\} \text{ infinite orthogonal, } p \wedge a_n = 0 \ (\forall n), \quad p \not\leq a_n \ (\forall n), \quad (2)$$

and $(\star)$ is satisfied *trivially* by the plain product $\prod_n \text{MO}_2$: take $a_n = a$ in coordinate n (zero elsewhere), and $p = \bigvee_n b_n$ where $b_n = b$ in coordinate n (zero elsewhere) — a genuine countable *orthogonal* join, so $p = (b, b, b, \dots)$ exists by σ -orthocompleteness. Then $p \wedge a_n = 0$ (coordinate n : $a \wedge b = 0$ in MO_2 ; elsewhere $a_n = 0$) and $p \not\leq a_n$ (coordinate n : $b \not\leq a^\perp$). Since $\prod_n \text{MO}_2$ is concrete,

σ -complete (a product of finite, hence complete, lattices), non-Boolean, and infinite, it satisfies $(\star)$ and *every* combinatorial hypothesis one might impose — yet it is exactly the *segregated* object the open problem must exclude. Call an OML *segregated* when its non-distributivity is confined to a direct product (or central decomposition) of finite blocks, so that every dispersion-free state factors through the blocks and the infinitary structure carries no contextuality of its own. So the combinatorial axioms (σ -completeness, concreteness, $(\star)$) are *not* the discriminator; the missing ingredient is *non-segregation*, a structural rather than a closure property.

Remark 5.5 (Absence of two-valued homomorphisms; the Boolean boundary). The universal closure above (a faithful set representation forces distributivity) already rules out the point-based route for every non-Boolean OML. The witness $\prod_n \text{MO}_2$ shows the failure concretely and at the level of points: MO_2 admits *no* two-valued *homomorphism* — each of its four separating two-valued *states* ($a \mapsto 0, b \mapsto 0$, etc.) fails $\omega(a \vee b) \leq \omega(a) + \omega(b)$ since $a \vee b = \mathbf{1}$, so none preserves $\vee$ — and $\prod_n \text{MO}_2$ inherits this, since its diagonal copy $\{\mathbf{0}, \mathbf{1}, (a, a, \dots), (a^\perp, \dots), (b, b, \dots), (b^\perp, \dots)\}$ is a sub- MO_2 on which any homomorphism would restrict to one of MO_2 's (none exists). (This is special to MO_2 and its products — a generic non-Boolean OML such as $\mathbf{2} \times \text{MO}_2$ *does* carry homomorphisms, e.g. the projection; the universal obstruction is the distributivity argument, not homomorphism-scarcity.) So the witness is concrete (Definition 2.7: order-determining two-valued *states*) and σ -complete yet has no points in the homomorphism sense at all — the point-free target is forced, not engineered. The same arithmetic records the sharp Boolean boundary: an OML is Boolean iff it admits a *unital* set of *subadditive* states — one with, for each nonzero a , a subadditive state giving a value 1 (Pták–Pulmannová [35, Thm 1], for OMLs; the orthomodular-poset version fails). A concrete non-Boolean OML therefore cannot have its separating two-valued states form such a unital set: some fail subadditivity (here all four of MO_2 's fail $\omega(a \vee b) \leq \omega(a) + \omega(b)$). The forcing property to Boolean is thus subadditivity — a finitary condition — not σ -additivity.

The lesson survives elaboration. Navara's construction [32] — once its stateless Greechie block (the source of non-concreteness) is replaced by MO_2 — yields a concrete, σ -orthocomplete, non-Boolean OML carrying $(\star)$; yet it is $\prod_n \text{MO}_2$ with scaffolding, segregated in the same way, with no σ -tribe representation. So $(\star)$ and concreteness together are easy and do *not* reach the open problem: the missing ingredient is non-segregation, and that is what the witness must supply.

The known results constrain the open (σ -complete) problem from both sides without closing it: *no* impossibility theorem covers the concrete class (Remark 4.1), while the two *theorem-closed* routes to an engine — RDP and MacNeille (§4.3(i),(ii)) — are ruled out, leaving only the third, σ -Stone duality, which is itself open. What is required is not a new idea in the abstract but a representation in that gap: non-distributive enough to evade routes (i),(ii), yet σ -faithful where the known dualities are merely finitary.

Remark 5.6 (Why $L(H)$ +Gleason does *not* already settle this — the (A)/(B) point-space equivocation). “Probability from relations, point-free” may *look* delivered by $L(H)$ +Gleason. It is not: the appearance rests on conflating two point-spaces, and *realisation* is defined as concentration on the second. The two are (A) the Hilbert rays of H , which Gleason *requires* — the density ρ in $s = \text{tr}(\rho \cdot)$ is built from them — and (B) the McDonald–Bimbó dual filters $P(A)$, which Gleason *removes*, but only by being a representation theorem. Gleason is thus the *most* point-ful result available (every lattice state corresponds to the point datum ρ). What $L(H)$ exhibits is a *separation*, not point-freeness: σ -additive measures exist on the lattice (Gleason, for normal states), yet on the dual $S_0(L(H))$ no state extends to a charge at all (Proposition 3.2), so a fortiori none concentrates on $P(A)$. It cannot serve as the existence proof required, so the point-free σ -additive structure of §5 is the only candidate engine.

5.3. Two axes of obstruction. The recurring confusion of this subject (Remarks 5.5, 5.6) is best held as a 2×2 table on two *independent* axes, which must not be collapsed into one.

- **Axis 1 — distributivity = commutativity.** One condition in two dialects: a family of projections commutes *iff* the lattice they generate is distributive. “Commutativity” is the operator word, “distributivity” the lattice word, for the *same* dividing line — not different sides.
- **Axis 2 — concreteness.** A non-Boolean lattice can be *point-rich in rays* (atoms = 1-dim subspaces) yet *point-free in valuations* (no two-valued homomorphisms, Kochen–Specker), or the reverse: the two senses of “point” come apart (the (A)/(B) equivocation, Rmk 5.6).

	distributive (commuting)	non-distributive (non-commuting)
concrete (valuations span)	classical: Boolean σ -algebra, Loomis–Sikorski free	the descent witness lives here: a concrete σ -complete non-Boolean OML ($\prod_n \text{MO}_2$ lies here, but segregated)
non-concrete (KS: few valuations)	atomless measure algebra (valuation-poor)	$L(H)$: rays abundant, valuations KS-barred — the operator / e.s.a. world, the excluded pole

There is not one “lattice $\leftrightarrow$ space” dictionary but two, one per row: *Birkhoff–von Neumann* (bottom row) sends elements to closed subspaces and atoms to **rays**, with $a \vee b$ the closed span — rich in ray-points, KS-barred from valuation-points, hence non-concrete; *Stone / concrete-logic* (Gudder, top row) sends elements to sets of outcomes and atoms to points of Ω — rich in valuation-points, hence concrete. The operator question and the descent question share Axis 1 but sit in opposite rows: spectral projections *force* $L(H)$ (bottom-right), whereas the descent witness must be top-right. So the descent problem is the one cell no operator algebra can occupy — why a witness must be a pasting / countable colimit of concrete blocks, not a Hilbert-space construction, and why $L(H)$ +Gleason cannot settle it. In this language the open problem (route (iii)) is exactly: *is there a logic $\leftrightarrow$ geometry dictionary for the top-right cell?* — and a σ -Loomis–Sikorski for concrete OMLs is that missing dictionary. Axes 1 and 2 are independent: non-distributivity does not entail non-concreteness, and a witness must be concrete and non-distributive simultaneously.

6. DIRECTIONS

With the extension axis closed on $L(H)$ (Proposition 3.2), the combinatorial route shown to deliver only segregated objects (§5.2), and the question reduced to a σ -essential contextual state (§5.1), the problem admits two mutually exclusive resolutions: construct such a state, or prove none exists. The second is already established in the Polish-representable case (Derr–Williamson, Remark 5.1), so both turn on a single feature — whether a witness can fail to admit a regular Borel point-realisation.

Exhibit a σ -essential witness:

Construct an infinite concrete OML L carrying a σ -additive state w that is contextual yet witnessed by no finite sub-OML. The object can be pinned down. Write, for a finite sub-OML F ,

$$A_F := \overline{\text{con v}}(S_{\text{df}}(F)), \quad B_F := \overline{\text{con v}}\{s|_F : s \in S_{\text{df}}^\sigma\},$$

the full local dispersion-free hull and the hull of restrictions of the *global* σ -states. The inclusion

$$B_F \subseteq A_F$$

always holds, since a restriction $s|_F$ of a global dispersion-free state is a dispersion-free state on F . The two defining properties of w now combine. Because w is contextual, $w \notin \overline{\text{con v}}(S_{\text{df}}^\sigma)$; this hull is compact, so by Hahn–Banach a finite-support functional separates w from it, and the elements in its support generate

a finite sub-OML F with

$$w|_F \notin B_F.$$

Because w is σ -essential, on that same F

$$w|_F \in A_F.$$

The two statements, with $B_F \subseteq A_F$, give $w|_F \in A_F \setminus B_F$, so

$$B_F \subsetneq A_F.$$

The upshot: contextuality is finitely *detected* — it shows at the single finite F — though no finite sub-OML is itself contextual, and the whole target reduces to this one *restriction gap*, the failure of the global σ -states to restrict onto the full local hull. The obstruction is therefore carried by the σ -structure, not by any finite block.

One necessary condition follows. The witness must be *off-centre* (irreducible, with non-central infinite blocks): a central decomposition would route the infinitary content into a Boolean centre, where σ -additivity concentrates the contextual states onto points and removes them — this is the $\prod_n \text{MO}_2$ mechanism (§5.2). The two obvious recipes both fail this test: a compactness-failure limit (Remark 5.4) collapses to non- σ -additivity, and a pasting or countable colimit of Wright-type blocks collapses to segregation.

Prove no such witness exists:

A spanning theorem: the restriction gap above is always empty (off a contextual finite sub-OML, Wright), so σ -essential relational probability is operationally invisible — present finitely or not at all. This is already established in the Polish-representable case (Derr–Williamson, Remark 5.1); a full result must extend it to the non-representable case, where their Theorem D.6 hypothesis is unavailable. The Boolean boundary (subadditivity of a unital set of states, Remark 5.5) is a natural lever, but cannot reuse the $\prod_n \text{MO}_2$ mechanism, vacuous off-centre.

This also settles a subsidiary question: whether a singular state might extend where no normal state does, which would give a first example of extension without concentration. Proposition 3.2 excludes it by a *finite* family ($n = 4$), so the singular case — which was expected to require infinitary methods — is a self-contained finitary matter, not governed by the finitary-to- σ passage that carries the open question.

APPENDIX A. GLOSSARY OF DEFINITIONS

A self-contained list of the notions used in the paper. Entries that restate a numbered definition in the body cross-reference it; the remaining entries are stated here. Throughout, L is an orthomodular lattice with orthocomplementation $a \mapsto a^\perp$, bottom $\mathbf{0}$, top $\mathbf{1}$.

Lattice notions.

Orthocomplemented lattice; orthomodular lattice (OML).:

A bounded lattice with an order-reversing, involutive complement $a^\perp$ satisfying $a \vee a^\perp = \mathbf{1}$, $a \wedge a^\perp = \mathbf{0}$ (Definition 2.1); it is *orthomodular* if $a \leq b \Rightarrow b = a \vee (b \wedge a^\perp)$ (Definition 2.4). A *Boolean algebra* is a distributive OML.

Orthogonal; meet-zero; incomparable.:

$a \perp b$ means $a \leq b^\perp$; *meet-zero* means $a \wedge b = \mathbf{0}$. Orthogonal implies meet-zero; the converse fails off the Boolean case, and that gap drives the subject (Definition 2.2).

Compatible; block.:

a, b are *compatible* if they generate a Boolean subalgebra, equivalently $a = (a \wedge b) \vee (a \wedge b^\perp)$. A *block* is a maximal pairwise-compatible subset of L ; it is a maximal Boolean sub-OML, and every OML is the union of its blocks. (In a concrete logic, compatibility of A, B is $A = (A \cap B) \cup (A \cap B^c)$.)

Completeness.:

L is σ -*complete* if every countable subset has a join; it is σ -*orthocomplete* if every countable *pairwise-orthogonal* family has a join (Definition 2.5).

Centre; irreducible.:

The *centre* $Z(L)$ is the set of elements compatible with all of L ; it is a Boolean subalgebra. L is *irreducible* if $Z(L) = \{\mathbf{0}, \mathbf{1}\}$ (no nontrivial direct-product decomposition).

 MO_n ; $L(H)$.:

MO_n is the modular ortholattice with $\mathbf{0}, \mathbf{1}$ and n complementary atom-pairs (MO_2 : atoms $a, a^\perp, b, b^\perp$, all distinct meets $\mathbf{0}$); it is non-distributive for $n \geq 2$. $L(H)$ is the lattice of closed subspaces of a Hilbert space H (orthocomplement = orthogonal complement, join = closed span), the prototype non-distributive OML.

Representations.**OMP; concrete logic; order-determining states.:**

An *orthomodular poset* (OMP) weakens an OML to a partial join (only of orthogonal elements). L is a *concrete logic* (set-representable) if it embeds in some $\mathcal{P}(\Omega)$ with $a^\perp \mapsto \Omega \setminus a$ and orthogonal joins as disjoint unions, equivalently if it carries an *order-determining* (separating) family of two-valued states (Definition 2.7).

Faithful set representation; σ -tribe.:

An injection $h : L \rightarrow \mathcal{P}(\Omega)$ with $h(a \wedge b) = h(a) \cap h(b)$, $h(a \vee b) = h(a) \cup h(b)$, $h(a^\perp) = \Omega \setminus h(a)$ (preserving *all* meets and joins as $\cap, \cup$); a σ -*tribe* / Loomis–Sikorski representation also preserves countable joins. By Theorem 5.2 only Boolean L admits one. (Contrast a concrete logic, which preserves only *orthogonal* joins as unions.)

Dual space $S_0(A)$; canonical points $P(A)$.:

$S_0(A)$ is the McDonald–Bimbó space of filters of A ; $P(A) = \{\uparrow p : p \in A\}$ is the set of principal filters, one per element (Definition 2.12).

States and measures.**State; meet-additive; charge.:**

A *state* is $s : L \rightarrow [0, 1]$ with $s(\mathbf{1}) = 1$ and $s(a \vee b) = s(a) + s(b)$ for $a \perp b$ (Definition 2.9); a *charge* is a finitely additive set function on a Boolean algebra (Definition 2.13).

 σ -additive state.:

A state s with $s(\bigvee_n a_n) = \sum_n s(a_n)$ for every countable pairwise-orthogonal $\{a_n\}$ (Definition 2.5).

Dispersion-free state.:

A two-valued state, i.e. a state taking only the values 0, 1 (equivalently a $\{0, 1\}$ -homomorphism on each block); S_{df} denotes their set and S_{df}^σ the σ -additive ones. These are the deterministic / hidden-variable points.

Normal; singular state on $L(H)$.:

s is *normal* if $s(\cdot) = \text{tr}(\rho \cdot)$ for a density operator ρ ; *singular* if s vanishes on every finite-rank projection (Definition 2.11).

Subadditive state.:

A state s with $s(a \vee b) \leq s(a) + s(b)$ for *all* a, b (not only orthogonal pairs). A family of states is *unital* if for each $a \neq \mathbf{0}$ some member gives $s(a) = 1$.

The descent vocabulary.

Contextual state.:

A state w that is *not* in the closed convex hull $\overline{\text{conv}}(S_{\text{df}})$ of the dispersion-free states — equivalently (Lemma 5.3) a state with no hidden realisation, i.e. not a mixture of dispersion-free states.

A non-contextual state is one that is such a mixture.

σ -essentially contextual state.:

A σ -additive state w on a σ -complete L that is *contextual* — $w \notin \overline{\text{conv}}(S_{\text{df}}^\sigma)$ — yet whose restriction to *every finite sub-OML* of L is non-contextual. (All three clauses are required: σ -additive, globally contextual, finitely non-contextual. The finite version without the third clause is already inhabited, Wright’s pentagon.)

Extension axis; descent axis.:

The *extension* axis asks whether a state extends to a charge on the full clopen algebra of $S_0(A)$ (a Horn–Tarski feasibility question); the *descent* axis asks whether such a charge concentrates on the physical points $P(A)$ (governed by σ -additivity). See §3.

Segregated.:

Of an OML: one whose non-distributivity is confined to finite blocks, so that $S_{\text{df}} = S_{\text{df}}^\sigma$ and no σ -essential leak is possible (e.g. $\prod_n \text{MO}_2$, §5.2).

Prior-art objects (§5, Remark 5.1).

Pre-Dynkin-system; Dynkin-system.:

In Derr–Williamson’s terminology, a *pre-Dynkin-system* on Ω is a concrete OMP (a family of subsets containing Ω , closed under complement and under disjoint unions); a *Dynkin-system* additionally is closed under countable disjoint unions — a concrete σ -complete OML.

σ -extendable.:

In Derr–Williamson, a countably additive probability μ_σ on a Dynkin-system $D_\sigma \subseteq \text{Borel}(\Omega)$ (Ω Polish) is σ -*extendable* if it extends to a countably additive probability on $\text{Borel}(\Omega)$.

Polish space; inner regular.:

A *Polish* space is a separable completely metrizable space. A measure μ is *inner regular* if $\mu(F) = \sup\{\mu(K) : K \subseteq F, K \text{ compact}\}$.

REFERENCES

- [1] Samson Abramsky and Adam Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13:113036, 2011.
- [2] Costantino Budroni and Giovanni Morchio. The extension problem for partial Boolean structures in quantum mechanics. *Journal of Mathematical Physics*, 51(12):122205, 2010. arXiv:1010.4662.
- [3] Leslie J. Bunce and J. D. Maitland Wright. The Mackey–Gleason problem. *Bulletin of the American Mathematical Society*, 26(2):288–293, 1992.
- [4] Dominika Burešová and Pavel Pták. On the set-representable orthomodular posets that are point-distinguishing. *International Journal of Theoretical Physics*, 62, 2023. Also arXiv:2401.13798.
- [5] Sarah Cannon. *The Spectral Presheaf of an Orthomodular Lattice: A Generalisation of Stone Duality to Orthomodular Lattices*. PhD thesis, University of Oxford, 2013. See also Cannon and Döring, “A generalisation of Stone duality to orthomodular lattices,” in *Contemporary Logic and Computing* (2018).
- [6] Emmanuel Chetcuti and Anatolij Dvurečenskij. On the Gleason theorem for effect algebras. *Mathematica Slovaca*, 55(3):297–308, 2005.
- [7] Thierry Coquand and Bas Spitters. Integrals and valuations. *Journal of Logic & Analysis*, 1(3):1–22, 2009.
- [8] Bruno de Finetti. *Theory of Probability*, volume 1. Wiley, 1974.
- [9] Rabanus Derr and Robert C. Williamson. Systems of precision: Coherent probabilities on pre-dynkin-systems and coherent previsions on linear subspaces. *arXiv preprint arXiv:2302.03522*, 2023.
- [10] Andreas Döring. Quantum states and measures on the spectral presheaf. *Advanced Science Letters*, 2(3):291–301, 2009. arXiv:0809.4847.
- [11] Anatolij Dvurečenskij. Loomis–sikorski theorem for σ -complete MV-algebras and ℓ -groups. *Journal of the Australian Mathematical Society*, 68(2):261–277, 2000.

- [12] Anatolij Dvurečenskij. Loomis–sikorski representation of monotone σ -complete effect algebras. *Fuzzy Sets and Systems*, 157(5):683–690, 2006.
- [13] Arthur Fine. Hidden variables, joint probability, and the Bell inequalities. *Physical Review Letters*, 48(5):291–295, 1982.
- [14] D. H. Fremlin. *Measure Theory, Vol. 3: Measure Algebras*. Torres Fremlin, Colchester, 2004.
- [15] Hector Freytes. An equational theory for σ -complete orthomodular lattices. *Soft Computing*, 24(14):10257–10264, 2020.
- [16] Andrew M. Gleason. Measures on the closed subspaces of a Hilbert space. *Journal of Mathematics and Mechanics*, 6(6):885–893, 1957.
- [17] Richard J. Greechie, David J. Foulis, and Sylvia Pulmannová. The center of an effect algebra. *Order*, 12(1):91–106, 1995.
- [18] Stanley P. Gudder. *Stochastic Methods in Quantum Mechanics*. North-Holland, New York, 1979.
- [19] John Harding. Orthomodular lattices whose MacNeille completions are not orthomodular. *Order*, 8(1):93–103, 1991.
- [20] Chris Heunen, Nicolaas P. Landsman, and Bas Spitters. A topos for algebraic quantum theory. *Communications in Mathematical Physics*, 291(1):63–110, 2009. arXiv:0709.4364.
- [21] Chris Heunen, Nicolaas P. Landsman, and Bas Spitters. Bohrification. In Hans Halvorson, editor, *Deep Beauty: Understanding the Quantum World through Mathematical Innovation*, pages 271–313. Cambridge University Press, 2011. arXiv:0909.3468.
- [22] Alfred Horn and Alfred Tarski. Measures in Boolean algebras. *Transactions of the American Mathematical Society*, 64(3):467–497, 1948.
- [23] Gudrun Kalmbach. *Orthomodular Lattices*. Academic Press, 1983.
- [24] Hans G. Kellerer. Verteilungsfunktionen mit gegebenen marginalverteilungen. *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete*, 3:247–270, 1964.
- [25] Alexander A. Klyachko, M. Ali Can, Sinem Binicioğlu, and Alexander S. Shumovsky. Simple test for hidden variables in spin-1 systems. *Physical Review Letters*, 101(2):020403, 2008.
- [26] Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17(1):59–87, 1967.
- [27] A. N. Kolmogorov. *Foundations of the Theory of Probability*. Chelsea, 1933.
- [28] Dorothy Maharam. Consistent extensions of linear functionals and of probability measures. In *Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability, Vol. II: Probability Theory*, pages 127–147. University of California Press, Berkeley, 1972.
- [29] Patrick S. Mahon. When does observational coherence determine probability? Preprint, 2026.
- [30] Joseph McDonald and Katalin Bimbó. A Stone-type duality for orthomodular lattices. *Mathematical Logic Quarterly*, 2023. arXiv:2208.07430.
- [31] V. Müller. Jauch–piron states on concrete quantum logics. *International Journal of Theoretical Physics*, 32(3):433–442, 1993.
- [32] Mirko Navara. Regularity and σ -additivity of states on quantum logics. *Proceedings of the American Mathematical Society*, 115(2):427–429, 1992.
- [33] Itamar Pitowsky. *Quantum Probability — Quantum Logic*, volume 321 of *Lecture Notes in Physics*. Springer, Berlin, 1989.
- [34] Pavel Pták. Exotic logics. *Colloquium Mathematicum*, 54(1):1–7, 1987.
- [35] Pavel Pták and Sylvia Pulmannová. A measure-theoretic characterization of Boolean algebras among orthomodular lattices. *Commentationes Mathematicae Universitatis Carolinae*, 35(1):205–208, 1994.
- [36] K. P. S. Bhaskara Rao and M. Bhaskara Rao. *Theory of Charges: A Study of Finitely Additive Measures*. Academic Press, New York, 1983.
- [37] Gottfried T. Rüttimann. Weakly purely finitely additive measures. *Canadian Journal of Mathematics*, 46(4):872–885, 1994.
- [38] Alex Simpson. Measure, randomness and sublocales. *Annals of Pure and Applied Logic*, 163(11):1642–1659, 2012.
- [39] Josef Tkadlec. Boolean orthoposets — concreteness and orthocompleteness. *Mathematica Bohemica*, 119(2):123–128, 1994.
- [40] Veeravalli S. Varadarajan. *Geometry of Quantum Theory*. Springer, New York, 2nd edition, 1985.
- [41] Steven Vickers. A localic theory of lower and upper integrals. *Mathematical Logic Quarterly*, 54(1):109–123, 2008.
- [42] Peter Walley. *Statistical Reasoning with Imprecise Probabilities*. Chapman and Hall, London, 1991.
- [43] Sander Wolters. A comparison of two topos-theoretic approaches to quantum theory. *Communications in Mathematical Physics*, 317(1):3–53, 2013. arXiv:1010.2031.
- [44] Ron Wright. The state of the pentagon: a nonclassical example. In A. R. Marlow, editor, *Mathematical Foundations of Quantum Theory*, pages 255–274. Academic Press, New York, 1978.